RUI PAN

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EDUCATION

University of Pennsylvania — **Philadelphia, PA**
M.S.E in Computer Graphics and Game Technology — **Aug 2026 - May 2028**

Cornell University — **Ithaca, NY**
B.S in Information Science | Minor in Game Design | **GPA:** 3.92/4.0 — **Aug 2023 - May 2026**

- **Relevant Coursework:** Computer Animation, Procedural Computer Graphics, Physically Based Realistic Rendering, Computer Graphics, Advanced Game Architecture and Design

TECHNICAL SKILLS

- **Programming Languages:** C++, C#, Java, GLSL/HLSL, Typescript
- **Engines & Frameworks:** Unity, Shader Graph, Unreal Engine Material Graph, Maya, Houdini
- **Technical Arts:** VFX, DCC Tools, Procedural Generation, asset pipelines, performance optimization, PBR

WORKING EXPERIENCE

Tencent Games — Lightspeed Studios — **Shenzhen, China**
Gameplay Engineer Intern — **Jun 2026 – Aug 2026**

- Built a data-driven particle system from scratch with structured VFX parameters supporting programmatic and AI-assisted effect authoring.
- Profiled and optimized CPU/GPU simulation and rendering, increasing **25K-particle performance from 9 to 60 FPS (6.7×)**.
- Maintained an AI-assisted game content pipeline, improving generated asset robustness and reliability before integration into game configurations.
- Developed a visualization tool for **13 workflow nodes**, exposing live file I/O and streamed agent logs to support pipeline inspection and debugging.

Development in Games Association (Cornell) — **Ithaca, NY**
Art Lead, Combat Design Lead — **Aug 2025 – May 2026**

- Led an **eight-member art team**, established visual guidelines and an asset production pipeline, and contributed to delivery of approximately 75% of the game's visual assets and combat VFX.
- Developed an automated asset pipeline using Python and image processing to convert raw artwork into animation-ready sprites, reducing processing time from approximately one hour per animation to seconds through batch automation.
- Collaborated with artists, designers, and engineers on **five boss encounters**, aligning combat effects and visual presentation with gameplay readability and performance needs.

PROJECT EXPERIENCE

Gnome More Shrooms: Mobile Farming + Tower Defense Game — **Ithaca, NY**
Technical Artist & Art Lead — **Jan 2026 - May 2026**

- Built an isometric rendering pipeline for 2D assets, supporting **5+ rendering layers, 1,000+ tiles**, and dynamic entities with correct depth sorting.
- Created a Data-driven particle framework with **3 emitter archetypes, 40+ parameters, and 20+ effects**; built a visual VFX editor that cut iteration time from **~30 minutes to under 5 minutes**.
- Improved runtime performance from **30 to 60 FPS** through draw-call optimization, texture atlasing, asset pipeline improvements, and render queue sorting.
- Developed a Procedural Terrain Generation algorithm using layered Perlin noise with visual parameter controls, plus **10+ Python scripts** for batch sprite conversion, resizing, and sprite sheet processing.

The Paige: PBR Path Tracer — **Ithaca, NY**
Graphics Engineer — **Jan 2026 - May 2026**

- Implemented Monte Carlo path tracing in Nori with MIS, microfacet and dielectric materials, thin-lens depth of field, and volumetric scattering.
- Built an SAH-based BVH and parallel construction pipeline, reducing BVH build time from **4.1 to 0.78 seconds**.
- Added texture-masked emissive lights, bloom, and an adaptive median denoiser to produce glowing glyphs and reduce fireflies in volumetric renders.